Guidance and Control of Unmanned Autonomous Vehicles

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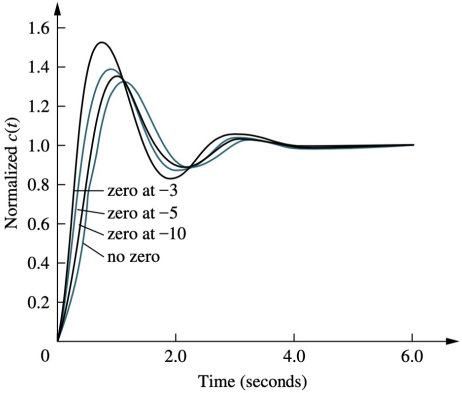
Unmanned Autonomous Vehicles: Feedback System with zeros in LHP

- What would happen if we add a zero in the LHP?
- Zeros affect the residue, or amplitude, of a response component, but do not affect the nature of the response.



- Increase in overshoot of response
- **Zero far from poles**: A simple gain factor and does not change the relative amplitudes of the components of the response.
- How to justify this observation?

Unmanned Autonomous Vehicles: Feedback System with zeros in LHP

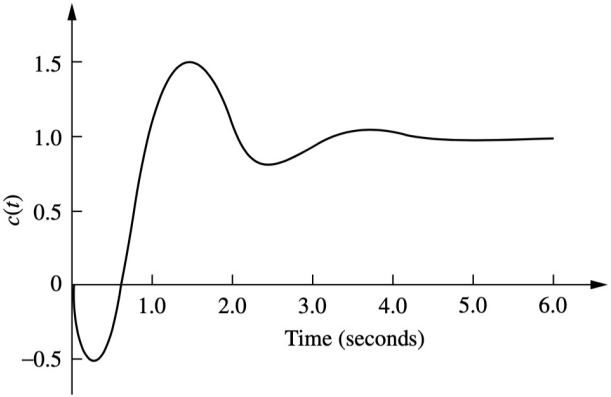
- Response with an added zero in the system transfer function:

$$\boxed{y(s) = (s+a)C(s) = sC(s) + aC(s)}$$

- A scaled version of the original response and its derivative
- If $a \gg 0$, the output is a scaled version of the original response. $y(t) \approx ac(t)$
- If a is not large, response contains the derivative of the original response.
- Lower the value of a, derivative term contributes more to the response and has a greater effect.
- For step responses, more overshoot in second-order systems because the derivative term will be additive around the first overshoot.
- Closer the zero is to the dominant poles, greater its effect on transient response.

Unmanned Autonomous Vehicles: Feedback System with zeros in RHP

- What would happen with addition of a RHP zero?
- Derivative term will be of opposite sign from the scaled response.



- Response begins to turn toward negative direction, even though its final value is positive.
- Nonminimum phase system

Unmanned Autonomous Vehicles: Feedback System design

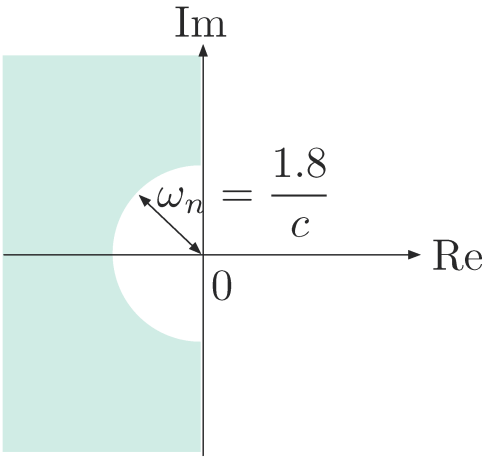
- How to handle the constraint on the rise time?
- Rise time:

$$T_r = \frac{1.8}{\omega_n}$$

- Suppose $T_r < c$ then what should be constraints on the natural frequency?

$$\omega_n > \frac{1.8}{c}$$

- What would be the feasible region in which poles lie?



Unmanned Autonomous Vehicles: Feedback System design

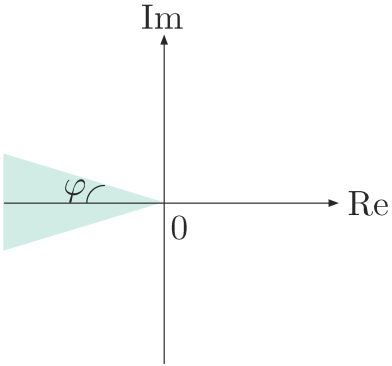
- How to handle the constraint on peak overshoot?
- Peak overshoot:

$$OS = e^{-\left(\pi \frac{\zeta}{\sqrt{1-\zeta^2}}\right)}$$

- Suppose $OS < c$, then what should be constraints on the damping ratio?

$$\frac{\zeta}{\sqrt{1-\zeta^2}} = \frac{\sigma}{\omega_d} = \cot\varphi, \;\; \sigma = \zeta\omega_n$$

- What would be the feasible region for the desired poles?

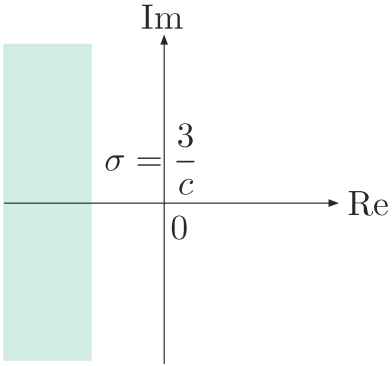


Unmanned Autonomous Vehicles: Feedback System design

- How to handle the constraint on settling time?
- Settling time: $T_s = \dfrac{4}{\zeta\omega_n} = \dfrac{4}{\sigma}$
- Suppose $T_s < c$, then what should be constraints on the real part of poles σ?

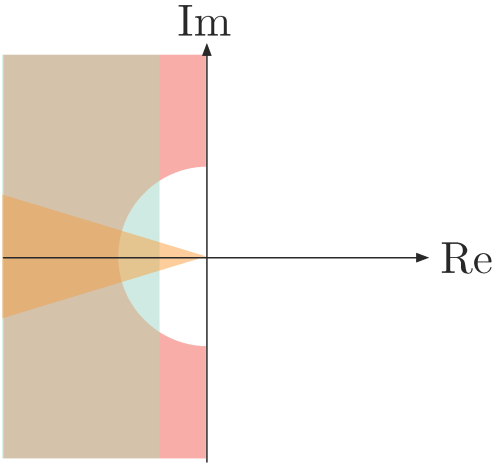
$$\frac{4}{\sigma} < T_s \implies \sigma > \frac{4}{\sigma}$$

- What would be the feasible region for the desired poles?



Deeper poles in the LHP $\implies$ Faster decay of transients, (smaller t_s)

How to handle constraints on all of these parameters for the closed-loop system?



References

- Norman S. Nise, "Control System Engineering", Wiley, Eighth Edition, 2020.

UAV Autopilot: Successive Loop Closure

Primary goal of autopilot

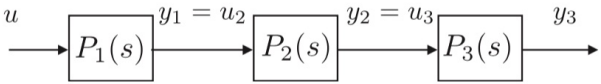
To control the inertial position and attitude of the vehicle (UAV)

- How to design the autopilot for the UAV?
- **Autopilot design**: Decoupled dynamics (longitudinal and lateral)
- A simplified problem with reduced order dynamics
- **Successive loop closure approach**

Successive loop closure

To close **several simple feedback loops in succession** around the open-loop plant dynamics rather than designing a single control system.

UAV Autopilot: Successive Loop Closure



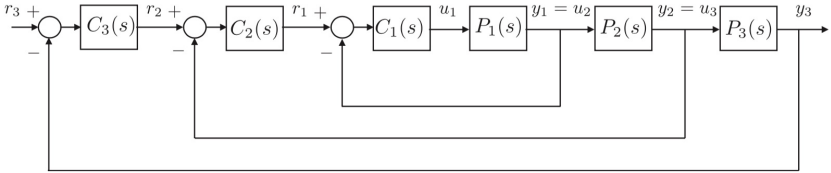
- Open loop dynamics

$$P(s) = P_1(s)P_2(s)P_3(s)$$

- $P_i(s)$: Lower (first or second) order transfer functions

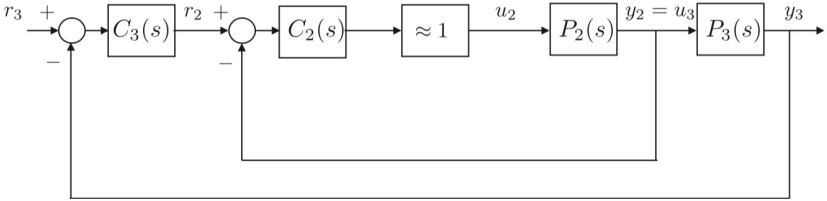
- Objective: To control y_3
- **Closed-loop control**: Feedback through output y_3
- Closure of feedback loops around all y_1, y_2, y_3, instead of only y_3
- **Necessary condition**: Inner loop should have the highest bandwidth, with the bandwidth of each successive loop smaller, by a factor of 5-10 in frequency

UAV Autopilot: Successive Loop Closure



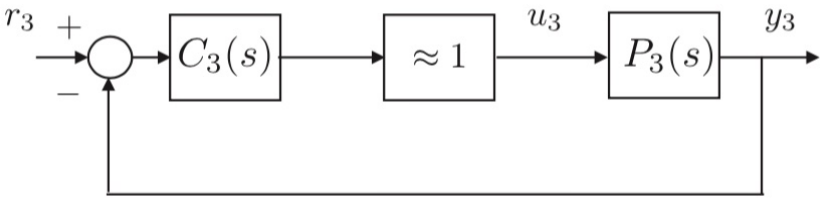
- **Compensators**: $C_i(s)$
- **Goal of compensator** $C_1(s)$: To design a closed-loop system from r_1 to y_1 having a bandwidth ω_{BW1}.
- For frequencies well below ω_{BW1}, closed-loop transfer function $\dfrac{y_1(s)}{r_1(s)} \approx 1$
- What would be the simplified model for the second loop?

UAV Autopilot: Successive Loop Closure



- **Critical step**: To design the next loop bandwidth, a factor of S_1 (5 to 10) smaller than the preceding loop.
- **Goal of compensator** $C_2(s)$: To design a closed-loop system from r_2 to y_2 having a bandwidth ω_{BW2}, with $\omega_{BW2} < \omega_{BW1}/S_1$.
- For frequencies well below ω_{BW2}, closed-loop transfer function $\dfrac{y_2(s)}{r_2(s)} \approx 1$

UAV Autopilot: Successive Loop Closure

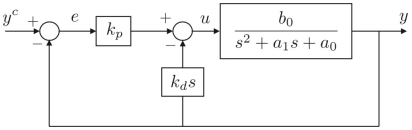


- **Goal of compensator** $C_3(s)$: To design a closed-loop system from r_3 to y_3 having a bandwidth ω_{BW3}, with $\omega_{BW3} < \omega_{BW2}/S_2$.
- Nature of plant models $P_1(s), P_2(s)$, and $P_3(s)$: First or second order
- **Conventional PID or lead-lag compensators**
- Transfer function-based design methods, such as root-locus approach

UAV Autopilot: Saturation Constraints and Performance

- **Successive loop closure**: Performance limitation of the system due to the inner-most loop
- What are other reasons for performance degradation?
  - Saturation constraints, such as bounded angular deflections
  - Physical bounds on ailerons restrict roll rate
  - Bound on total lateral acceleration due to aircraft stall at higher angle of attack
- **Goal of autopilot design**: To design bandwidth of the inner loop to be as large as possible, without violating the saturation constraints
- It allows outer loops to ensure **sufficient bandwidth separation** of the successive loops

UAV Autopilot: Saturation Constraints and Performance



- Consider a system $P(s) = \dfrac{b_0}{s^2 + a_1 s + a_0}$
- Closed-loop transfer function

$$\frac{y(s)}{y^c(s)} = \frac{b_0 k_p}{s^2 + (a_1 + b_0 k_d)s + (a_0 + b_0 k_p)}$$

- **Closed-loop poles**: Depends on gains
- **Actuator effort**: $\boxed{u = k_p e - k_d \dot{y}}$
- What terms affect the input demand or actuator effort?
- For $\dot{y} \approx 0$, size of error e and k_p decide about input u; $u \approx k_p e$
- For a stable system, largest control effort occurs immediately after step input
- Determination of proportional gain

$$\boxed{u^{\max} = k_p e^{\max} \implies k_p = \frac{u^{\max}}{e^{\max}}}$$

UAV Autopilot: Saturation Constraints and Performance

- Can we achieve any bandwidth at our choice? No
- How to compute the achievable bandwidth of a closed-loop system?
- Canonical form of second-order system, with no zeros

$$\frac{y(s)}{y^c(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Under-damped system

$$0 \leq \zeta < 1; \;\; \lambda = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

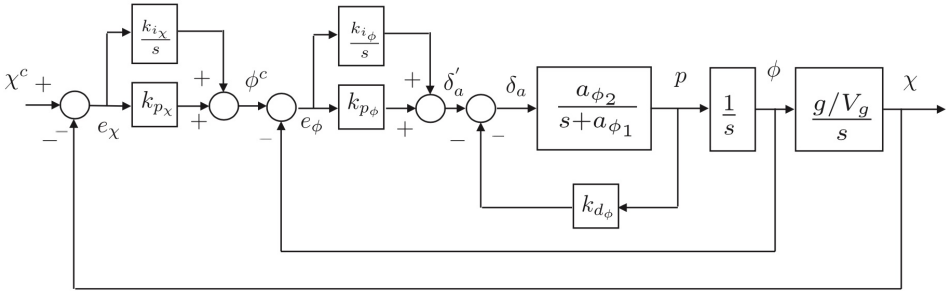
- Natural frequency of closed-loop system under saturation constraints

$$\omega_n = \sqrt{a_0 + b_0 k_p} = \sqrt{a_0 + b_0 \frac{u^{\max}}{e^{\max}}}$$

- **Upper limit on the bandwidth to avoid possible saturation on control input**

UAV Autopilot: Lateral directional Autopilot

- How to design the lateral autopilot of UAV?

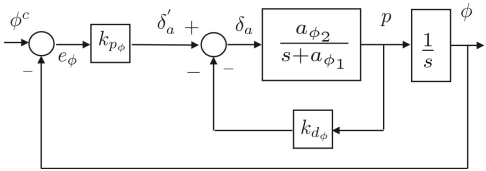


▷ Roll rate damping: k_{d_ϕ} ▷ Roll attitude: k_{p_ϕ}, k_{i_ϕ} ▷ Course angle: k_{p_χ}, k_{i_χ}

- How to select these gains?
- Successive selection of gains beginning with the inner loop and working outward.

UAV Lateral Autopilot: Roll Attitude Loop Design

- Linear design model: $\phi(s) = \dfrac{a_{\phi_2}}{s(s+a_{\phi_1})}\left[\delta_a(s) + \dfrac{1}{a_{\phi_2}} d_{\phi_2}(s)\right]$



- **Goal of inner loop**: To control roll angle and its rate
- **Gain design**: Based on desired response of closed-loop system
- Closed-loop transfer function

$$H_{\phi/\phi^c}(s) = \frac{k_{p_\phi} a_{\phi_2}}{s^2 + (a_{\phi_1} + a_{\phi_2} k_{d_\phi})s + k_{p_\phi} a_{\phi_2}}$$

UAV Lateral Autopilot: Roll Attitude Loop Design

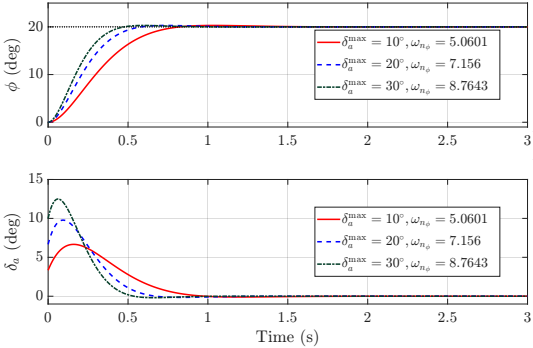
- Desired second-order closed-loop transfer function: $\dfrac{\phi^c(s)}{\phi(s)} = \dfrac{\omega_{n_\phi}^2}{s^2+2\zeta_\phi\omega_{n_\phi}s+\omega_{n_\phi}^2}$
- How to find this desired closed-loop transfer function?
- Equating coefficients: $\boxed{\omega_{n_\phi}^2 = k_{p_\phi}a_{\phi_2}, \quad 2\zeta_\phi\omega_{n_\phi} = a_{\phi_1} + a_{\phi_2}k_{d_\phi}}$
- Gain k_{p_ϕ} is chosen as per saturation bound on aileron deflections

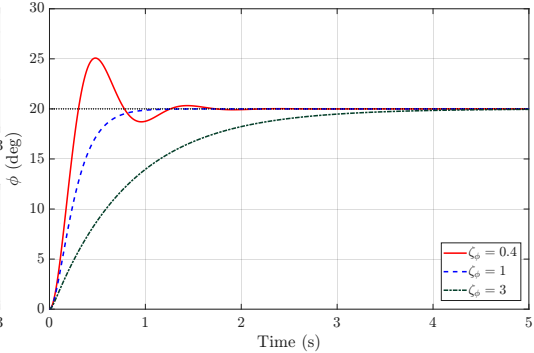
$$\boxed{k_{p_\phi} = \frac{\delta_a^{\max}}{e_{\phi^{\max}}}\text{sign}(a_{\phi_2}) \implies \omega_{n_\phi} = \sqrt{|a_{\phi_2}|\frac{\delta_a^{\max}}{e_{\phi^{\max}}}}}, \quad \boxed{k_{p_\phi} = \frac{\omega_{n_\phi}^2}{a_{\phi_2}} \text{ for unbounded } \delta_a}$$

- Gain k_{d_ϕ} based on gain k_{p_ϕ}, for chosen ζ_ϕ to satisfy other specifications

$$\boxed{k_{d_\phi} = \frac{2\zeta_\phi\omega_{n_\phi} - a_{\phi_1}}{a_{\phi_2}}}$$

UAV Lateral Autopilot: Roll Attitude response



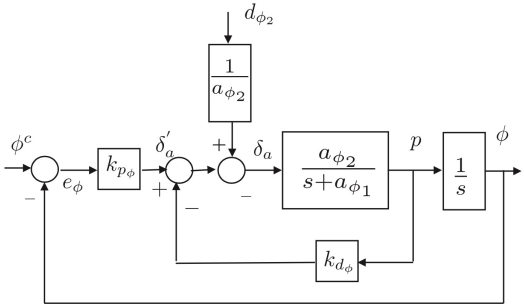


UAV Lateral Autopilot: Roll Attitude Loop Design Issues

- Zero steady-state tracking error in roll without integrator (Type-1 open loop TF)
- Recall the final value theorem and system type connection

- Presence of disturbance at the summing junction
- Neglected dynamics, wind gust, or turbulence

$$\phi(s) = \frac{a_{\phi_2}}{s(s+a_{\phi_1})}\left(\delta_a(s) + \frac{1}{a_{\phi_2}} d_{\phi_2}(s)\right)$$



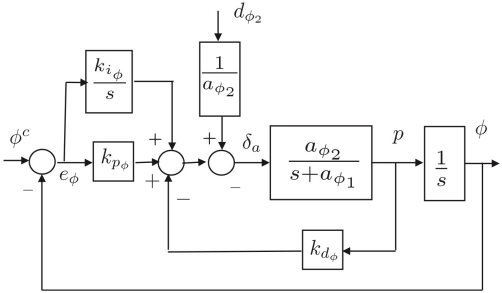
UAV Lateral Autopilot: Roll Attitude Loop Design Issues

- On solving for $\phi(s)$

$$\phi(s) = \left(\frac{1}{s^2 + (a_{\phi_1} + a_{\phi_2} k_{d_\phi})s + k_{p_\phi} a_{\phi_2}}\right) d_{\phi_2} + \left(\frac{a_{\phi_2} k_{d_\phi}}{s^2 + (a_{\phi_1} + a_{\phi_2} k_{d_\phi})s + k_{p_\phi} a_{\phi_2}}\right) \phi^c$$

- What would be the steady-state error for constant step disturbance $d_{\phi_2} = A/s$?
- Using the final-value theorem, steady-state error is $\dfrac{A}{a_{\phi_2} k_{p_\phi}}$, a nonzero value.
- For constant orbit, p, q, r and thus d_{ϕ_2} are constant.
- How to remove this steady-state error?
- To eliminate the steady-state error, an integrator is required.
- How to see if an integrator rejects the disturbances?

UAV Lateral Autopilot: Roll Attitude Loop Design using Integrator



$$\phi = \left[\frac{s}{s^3 + (a_{\phi_1} + a_{\phi_2}k_{d_\phi})s^2 + k_{p_\phi}a_{\phi_2}s + a_{\phi_2}k_{i_\phi}}\right] d_{\phi_2}$$
$$+ \left[\frac{a_{\phi_2}k_{p_\phi}\left\{s + (k_{i_\phi}/k_{p_\phi})\right\}}{s^3 + (a_{\phi_1} + a_{\phi_2}k_{d_\phi})s^2 + k_{p_\phi}a_{\phi_2}s + a_{\phi_2}k_{i_\phi}}\right] \phi^c$$

- **Steady-state error**:
  - Zero for constant d_{ϕ_2}
  - $\frac{A}{a_{\phi_2}k_{i_\phi}}$ for the ramp inputs
- Gain k_{i_ϕ} can be chosen using the root-locus technique
- How to select the gain k_{i_ϕ} using the root locus approach?

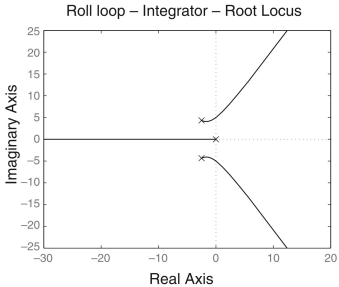
UAV Lateral Autopilot: Roll Attitude Loop Design using Integrator

- Characteristic equation of closed-loop system

$$s^3 + (a_{\phi_1} + a_{\phi_2} k_{d_\phi}) s^2 + k_{p_\phi} a_{\phi_2} s + a_{\phi_2} k_{i_\phi} = 0$$

- How to draw the root locus for above equation?

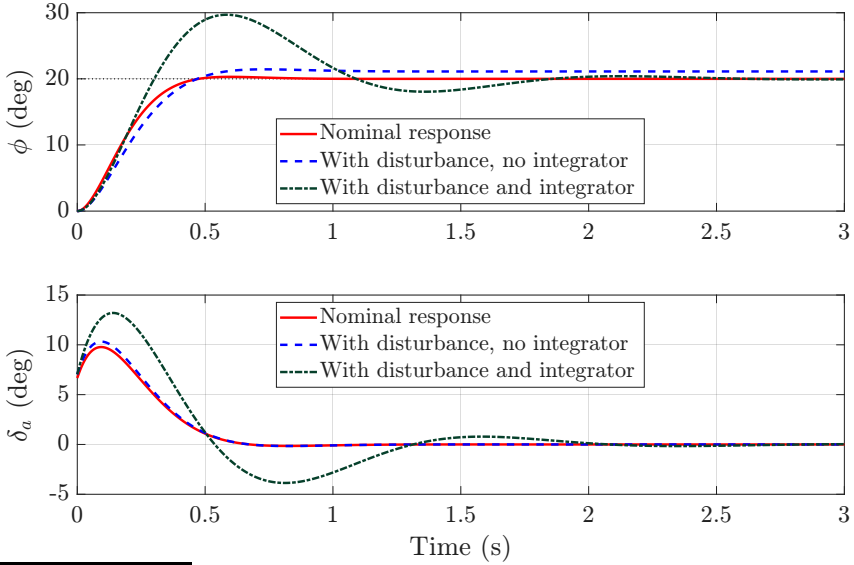
$$1 + k_{i_\phi} \left[\frac{a_{\phi_2}}{s\left(s^2 + (a_{\phi_1} + a_{\phi_2} k_{d_\phi}) s + k_{p_\phi} a_{\phi_2}\right)} \right] = 0$$



$$\boxed{\delta_a = k_{p_\phi}(\phi^c - \phi) + \frac{k_{i_\phi}}{s}(\phi^c - \phi) - k_{d_\phi} p}$$

Proper gain selection for better performance

UAV Lateral Autopilot: Roll Attitude Response Comparison



Textbooks/References

References

- Randal Beard and Timothy W. McLain, "Small Unmanned Aircraft: Theory and Practice", Princeton University Press, 2012

Thank you for your attention !!!